

3.22 Public key accelerator (PKA)

The public key accelerator (PKA) verifies ECDSA signatures, with all required computations performed within the accelerator. The application CPU manages the inputs and the outputs of the operation.

The main features of the PKA are:

- ECDSA signature verification
- Capability to handle operands up to 640 bits
- AMBA® AHB slave peripheral is accessible through 32-bit word single accesses only; otherwise an AHB bus error is generated, and write accesses are ignored

3.23 Timers and watchdogs

The devices include two advanced control timers, up to eleven general-purpose timers, two basic timers, two low-power timers, two watchdog timers, and one SysTick timer.

The table below compares the features of the advanced control, general-purpose and basic timers.

Table 5. Timer feature comparison

Timer type	Timer	Counter resolution	Counter type	Prescaler factor	DMA request generation	Capture / compare channels	Complementary outputs
Advanced control	TIM1, TIM8	16 bits	Up, down, Up/down	Any integer between 1 and 65536	Yes	4	4
General-purpose	TIM2, TIM5	32 bits	Up, down, Up/down	Any integer between 1 and 65536	Yes	4	No
	TIM12	16 bits	Up		No	2	No
	TIM15	16 bits	Up, down, Up/down		Yes	2	1
	TIM16, TIM17	16 bits	Up, down, Up/down			1	No
Basic	TIM6, TIM7	16 bits	Up	Any integer between 1 and 65536	Yes	0	No

3.23.1 Advanced-control timers (TIM1/TIM8)

The advanced-control timers (TIM1/TIM8) consist of a 16-bit autoreload counter driven by a programmable prescaler.

They may be used for various purposes, including measuring the pulse lengths of input signals (input capture) or generating output waveforms (output compare, PWM, complementary PWM with dead-time insertion).

Pulse lengths and waveform periods can be modulated from a few microseconds to several milliseconds using the timer prescaler and the RCC clock controller prescalers.

TIM1/TIM8 timer features include: